

EDUCATION

PhD Statistics , University of Edinburgh, United Kingdom	<i>Sep. 2022 – Aug. 2026</i>
Supervisors: Dr. Ioannis Papastathopoulos, Prof. Gabriele C. Hegerl	
Thesis: <i>Geometric considerations for generative modelling of multivariate extremes</i>	
Visiting PhD student , KAUST, Saudi Arabia	<i>Feb. 2024 – May 2024 and Nov. 2025</i>
Supervisor: Prof. Raphaël Huser	
MSc Mathematics and Statistics , McGill University, Canada	<i>Sep. 2020 – June 2022</i>
Supervisor: Prof. Christian Genest	
Thesis: <i>A spatial Gamma-Gamma model for the statistical postprocessing of ensemble weather forecasts</i>	
BSc Mathematics and Computer Science , McGill University, Canada	<i>Sep. 2017 – May 2020</i>

RESEARCH INTERESTS

- Extreme value theory, Multivariate and spatial statistics, Probabilistic forecasting
- Machine learning, generative modelling
- Environmental applications in the context of climate change.

PROFESSIONAL ACTIVITIES

Hydro-Québec Research Institute (IREQ)	<i>Jan. 2022 – Aug. 2022</i>
- Research partnership to develop methods for the statistical post-treatment of weather forecasts, specifically of extreme meteorological events.	
DataHub, National Bank of Canada	<i>May 2019 - Sept. 2020</i>
<i>Data Science Internship</i>	
- Development and automation of the official DataHub credit card data base.	
- Statistical modeling of credit card use and behaviors.	

RELEVANT EXPERIENCE

Postgraduate representative, School of Mathematics (University of Edinburgh)	<i>Sept. 2024 – Oct. 2025</i>
Representative of the postgraduate students of the <i>School of Mathematics</i>	
- Intermediary between the students and the School administration.	
- Organisation of social events within the School.	
- Representing the <i>School of Mathematics</i> at the <i>Martingale PhD Navigator Programme</i> events for future scholars (December 2024 and 2025).	
Teaching Assistantships (University of Edinburgh)	
Tutoring workshops. Marking of assignments and examinations. List of courses:	
- MATH10093 – Statistical Computing	<i>Jan. – May 2023, Jan. – May 2025, and Jan. – May 2026</i>
- MATH08065 – Computing and Numerics	<i>Jan. – May 2026</i>
- MATH10096 – Applied Statistics	<i>Jan. – May 2025 and Jan. – May 2026</i>
- MATH11199 – Python Programming	<i>Sep. – Dec. 2025</i>
- MATH08066 – Probability	<i>Sep. – Dec. 2023 and Sep. – Dec. 2025</i>
- MATH10064 – Multivariate Data Analysis	<i>Jan. – May 2025</i>
- MATH11176 – Extended Statistical Programming	<i>Sep. – Dec. 2023</i>
- MATH11187 – Generalised Regression Models	<i>Sep. – Dec. 2023</i>
- MATH08051 – Statistics	<i>Jan. – May 2023</i>

Teaching Assistantships (McGill University)

Preparation of tutorial sessions, office hours availabilities for course related questions. Marking of midterm and final exams, supervision of assignment marking. List of courses:

- MATH 324 – Statistics *Jan. 2022 – May 2022*
- MATH 323 – Probability *Sep. 2021 – Dec. 2021*
- MATH 203 – Principles of Statistics *Jan. 2021 – May 2021*

Statistics Helpdesk (McGill University)

Sep. 2020 – Dec. 2021

Providing help to undergraduate students in Statistics and Probability. Theoretical and programming guidance for courses in the range MATH 2XX - 5XX.

Undergraduate Project Guidance (McGill University)

Guidance of three students towards the completion of an undergraduate project in extreme value theory.

- MATH 470 – Honours Research Project *Jan. 2022 – May 2022*
- MATH 410 – Majors Project *Jan. 2021 – May 2021 and Sep. 2021 – Dec. 2021*

SOFTWARE

- Python code [geometricExtremesNF](#): Creator and maintainer. Based on De Monte et al. (2025)
- R package [geometricExtremes](#): Creator and maintainer. Based on Papastathopoulos et al. (2023).
- R code [TSExtremes](#): Collaborator. Based on Auld et al. (2026).

AWARDS

School of Mathematics, University of Edinburgh

Sep. 2023

- Full PhD Studentship: £27,600 tuition and £20,800 stipend per year

Mitacs Canada

Jan. 2022

- Funding for research on extreme value modeling in partnership with Hydro-Québec. \$30,000 (CAD)

SCIENTIFIC ARTICLE REVIEWING

Journals: Statistics and Computing.

PROGRAMMING

Languages: R, Python, Julia, Bash, SQL, LaTeX.

High-performance computing: Eddie (University of Edinburgh), Ibex (King Abdullah University of Science and Technology), CASIR (Hydro-Québec).

Cloud computing: Microsoft Azure (National Bank of Canada).

LANGUAGES

Bilingual: English, French.

PRESENTATIONS AND POSTER SESSIONS

Invited

- Generative AI Modelling for Extreme Events (GAME), Bologna, Italy: *Multivariate extremal index and clustering in multivariate extremes: A case study of Dutch hydrological extremes.* *Jun. 2026*
- CfS Annual Conference 2026, Edinburgh, United Kingdom: *Multivariate extremal index and clustering in multivariate extremes: A case study of Dutch hydrological extremes.* *Jun. 2026*
- Generative AI Modelling for Extreme Events, University of Edinburgh Generative AI Laboratory, Edinburgh, United Kingdom: *Generative modelling of multivariate geometric extremes using normalising flows.* *Jun. 2025*
- Statistics seminar, University of Exeter, Exeter, United Kingdom: *Geometric extreme value theory – A normalising flows approach.* *Feb. 2025*
- CMStatistics, London, United Kingdom: *Geometric extreme value theory – A normalising flows approach.* *Dec. 2024*
- CMStatistics, Berlin, Germany: *Bayesian inference for radially-stable distributions.* *Dec. 2023*
- BIRS-IMAG Modern Statistical and Machine Learning Approaches for High-Dimensional Compound Spatial Extremes, Granada, Spain (short presentation): *Geometric inference for Hüsler–Reiss random vectors.* *May 2023*

- Séminaires de sciences de la décision HEC Montréal, Montréal, Canada: *Multivariate extremes – A geometric Bayesian inference approach*. Sep. 2023

Contributed

- 14th International Conference on Extreme Value Analysis, University of North Carolina Chapel Hill, United States: *Generative modelling of multivariate geometric extremes using normalising flows* Jun. 2025
- 15th International Meeting on Statistical Climatology, Toulouse, France: *Flood risk modelling using geometric extreme value theory*. Jun. 2024
- CfS Annual Conference 2024, Edinburgh, United Kingdom (poster): *Multivariate radial Pareto distributions: a geometric approach to the statistical modelling of multivariate extremes*. Jun. 2024
- 13th International Conference on Extreme Value Analysis, Milan, Italy (poster): *Bayesian approach to geometric inference for multivariate extremes*. Jun. 2023

PUBLICATIONS

Research

- De Monte, L. & Papastathopoulos, I. (2026), *A directional Poisson point process for univariate extremes*, [Statistics & Probability Letters](#) **235**, 110722.
- De Monte, L., Huser, R., Papastathopoulos, I., and Richards, J. (2025) *Generative modelling of multivariate geometric extremes using normalising flows*, [arXiv](#) preprint.
- Papastathopoulos, I., De Monte, L., Campbell, R. & Rue, H. (2023) *Statistical inference for radially-stable generalized pareto distributions and return level-sets in geometric extremes*, [arXiv](#) preprint.

Book chapter

- Auld, G., De Monte, L. & Papastathopoulos, I. (2026) Time series in extremes. In de Carvalho, M., Huser, R., Naveau, P., and Reich, B. J., editors, *Handbook on Statistics of Extremes*. Chapman & Hall/CRC, Boca Raton, FL.

Other

- Richards, J., De Monte, L. Review of “Risk Revealed: Cautionary Tales, Understanding and Communication” by Paul Embrechts, Marius Hofert, and Valérie Chavez-Demoulin. [JABES](#) (2024).
- Genest, C. and De Monte, L. (2021) *À la recherche de la Rondurie*, [Accromath](#), Volume 16 p.24-29.

Thesis

- De Monte, L. *A Spatial Gamma-Gamma Model for the Statistical Postprocessing of Ensemble Weather Forecasts*. Master’s thesis. [McGill University](#), 2022.